

# PAUL MITCHELL

## TECHNOLOGY CONSULTING | SYSTEMS ARCHITECTURE & RISK ADVISORY

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### PROFESSIONAL SUMMARY

William & Mary Computer Science M.S. candidate (3.95 GPA, AI/ML & Systems) and B.S. graduate combining hands-on Linux systems administration (100+ instructional machines), cloud telemetry pipelines, and quantitative data analytics. Experienced in translating complex technical environments, security controls, and infrastructure trade-offs into actionable client solutions, executive dashboards, and audit-ready operational frameworks. Proven ability to bridge deep engineering rigor with strategic business decision-making.

### EDUCATION

#### College of William & Mary

Williamsburg, VA

Master of Science in Computer Science (AI/ML & Systems Concentration) | Expected Dec 2026 | GPA: 3.95

Bachelor of Science in Computer Science | Graduated Dec 2024

Relevant Coursework: Advanced Software Engineering, Machine Learning Systems, Deep Learning, Data Mining

### TECHNICAL SKILLS

Systems & Infrastructure: Linux (Ubuntu/RHEL), Windows, SSH, DNS, static IPs, Netplan, NFS, Docker, Puppet, system hardening

Monitoring & Operations: Prometheus, Grafana, dashboard engineering, alert rule design, log auditing, capacity planning

Programming & Scripting: Python, Bash, PowerShell, Java, SQL, YAML, Git/GitLab CI, FastAPI, RESTful APIs

Advisory & Analytics: Risk assessment, technical runbooks, anomaly detection, threshold calibration, root-cause analysis, Excel

### TECHNICAL & PROFESSIONAL EXPERIENCE

#### IT Administrator | William & Mary Technology Support Center | Williamsburg, VA

Feb 2025 - Present

- Administer, troubleshoot, and support more than 100 Linux instructional systems, resolving network connectivity, routing, DNS, account, and software issues for campus users.
- Built Prometheus and Grafana telemetry dashboards tracking GPU compute clusters, storage health, and node uptime, accelerating incident response and issue resolution.
- Conducted architectural evaluation of multi-user Docker containerization, documenting security isolation, resource overhead, and operational maintenance trade-offs for leadership.
- Standardized network configurations and maintained structured inventory records in Excel, eliminating configuration drift across multi-room compute environments.
- Developed automated Bash recovery scripts for stalled network daemons, eliminating recurring manual administrator interventions.

#### Data Analytics & Engineering Extern | Extern, Amazon-Sponsored Project | Remote

Jun 2025 - Sep 2025

- Engineered reproducible Python and AWS S3 data pipelines to clean and structure ~20,000 records, cutting recurring data prep cycles from hours to minutes.
- Built an interactive Streamlit analytics application that synthesized complex model outcomes into actionable executive decision-support metrics for program mentors.
- Evaluated predictive modeling trade-offs using LightGBM and NLP sentiment clustering, presenting strategic recommendations directly to stakeholders.
- Automated dataset versioning and storage reconciliation using DVC on AWS S3 to maintain audit-ready experimental lineages.

### KEY TECHNICAL & SYSTEMS PROJECTS

#### Real-Time AI Security & Prompt Injection Defense Proxy | Python, FastAPI

2026

- Architected pre-inference proxy middleware analyzing token structures to intercept adversarial prompts before model execution; benchmarked scoring latency at 0.038ms to 1.11ms.
- Documented enterprise integration patterns and security trade-offs for production LLM deployments, eliminating need for secondary verification model calls.

#### Enterprise IT Asset & Security Configuration Tracker | Python, Excel

2026

- Built an asset governance system tracking configuration drift, documentation gaps, and patch compliance across 100+ production instructional hosts.
- Designed automated normalization workflows for hardware lifecycles and software licenses, establishing audit-ready record maintenance procedures.

#### Healthcare Network Anomaly Detection Engine | Python, scikit-learn

2025

- Designed network telemetry inspection workflows featuring automated feature extraction and threshold calibration to support SOC operational alerting and threat triage.
- Translated complex traffic variance models into operational runbooks and decision matrices for technical support personnel.